

CEREBRO-X

Computational Drug-DDS Engineering Report

Field	Value
Drug Name	Lecanemab
Molecular Class	monoclonal_antibody
Molecular Weight	13060.3 Da
Recommended DDS	Tf-PEG-Liposome
Carrier Type	liposome
Surface Ligand	Transferrin
BBB Enhancement	N/A%
Composite Score	100.0/100
Disease Indication	CNS
Report Generated	2026-07-25 06:16 UTC
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Modules Completed	50/62 science modules

■ This report is generated from computational models. All predictions must be validated experimentally before clinical application.

1. Drug Molecular Profile

Property	Value	Source / Method
Name	Lecanemab	Input
MW (Da)	13060.3	ChEMBL / UniProt
LogP	-0.15	ADMET predictor
Half-life (days)	N/A	PubMed NLP
Protein Binding (%)	N/A	Calculated
BBB Penetration (%)	77.01	Pardridge 2012
Molecule Class	monoclonal_antibody	Input / inferred
Aqueous Solubility	N/A	ADMET
P-gp Substrate	Unknown	QSAR panel
ADMET Score	N/A	Multi-endpoint
ADMET Grade	N/A	A-F scale

1a. Drug Delivery Barriers Identified (Auto-detected)

The pipeline automatically identified 1 delivery barriers based on molecular data. Each is resolved by the recommended DDS.

[CRITICAL] Large molecular weight -- passive BBB diffusion impossible	
Evidence:	MW = 13.1 kDa (Lipinski Rule 5 limit: 500 Da for passive)
Without DDS:	0% brain penetration -- purely physical barrier
With DDS:	Active transcytosis delivers 10.0% payload to CNS
DDS Solution:	liposome encapsulation (75% EE)
Scientific basis:	BBB tight junctions (pore ~1nm) physically block molecules >500Da. Encapsulation in liposome (105nm) enables

2. DDS Ranking — Composite Score Methodology

All DDS formulations ranked by Composite Score = weighted combination of DLVO colloidal stability, transcytosis driving force, endosomal escape, stealth index, BBB receptor affinity, and CNS bioavailability. Scores are physics-derived — not estimated.

DDS Name	Carrier	Ligand	Score	DLVO kT
Tf-PEG-Liposome	liposome	Transferrin	100.00	8.90
RVG29-Liposome-pH	liposome	RVG29	100.00	14.84
ApoE-LNP	lnp	ApoE	100.00	-6.24
Tf-SLN-Stealth	solid_lipid	Transferrin	100.00	-0.12
RVG29-PLGA-NP	plga	RVG29	100.00	5.83
Lf-Polymer-Micelle	micelle	Lactoferrin	97.00	-4.06
Bare-PLGA	plga	None	78.00	-16.16
AAV9-vector	aav9	None	53.00	-2.37

2a. Recommended DDS — Detailed Profile

Parameter	Value	Biophysical Meaning
Composite Score	100.0	Overall Drug+DDS suitability (0-100)
BBB Enhancement	N/A%	% of dose crossing BBB via receptor transcytosis
CNS Bioavailability	N/A%	% dose reaching brain as free drug
DLVO Stability	8.9 kT	>25kT = colloiddally stable in blood
Endosomal Escape	N/A	Fraction escaping lysosomes (0-1)
Stealth Index	N/A	MPS evasion (0=opsonised, 1=invisible)
Protein Corona	N/A nm	Corona thickness; thicker = faster clearance
MPS Clearance	N/A h	Hours before liver/spleen removes DDS
Payload Efficiency	N/A%	% dose delivered as active drug to CNS
CARPA Risk	0.53	Complement activation risk (0=safe)

3. PBPK-CNS Digital Twin — 6-Compartment Simulation

Physiologically Based Pharmacokinetic model solved with scipy Radau ODE integrator (stiff system). Compartments: Plasma, BBB endothelium, Brain ISF, Brain cells, CSF, Peripheral. All parameters from molecular data — not estimated. Reference: Pardridge 2012; Bhatt 2013.

PK Metric	Value	Compartment	Source
Model	BiologicPBPK (FcRn + CNS transcytosis) T _{1/2} =470.4h		Shah DK & Betts AM (2012) JPKPD 39:67-86; Pardridge 2012
C _{max} (brain)	0.006286 µg/mL	Brain/CNS	Peak brain concentration
AUC (brain)	1.50098 µg·h/mL	Brain/CNS	Total brain exposure
AUC (plasma)	682.320 µg·h/mL	Plasma	Systemic exposure
K _{p,brain}	0.00220	Brain/Plasma	Partition coefficient
BBB penetration	0.2200%	BBB	BiologicPBPK (two-compartment + FcRn + CNS transcytosis)
K _{p,uu brain}	N/A	Unbound	Unbound K _p (pharmacodynamically relevant)
t _{_above 10%C_{max}}	N/A h	Brain ISF	Duration above therapeutic threshold
Glymphatic t _{1/2}	N/A h	Brain	Clearance half-life in brain
BBB PS _{_in}	N/A mL/h	BBB	Permeability-surface area product
BBB PS _{_out}	N/A mL/h	BBB	Efflux transport rate
Disease state	healthy	BBB	BBB integrity modifier applied

4. In-Silico Drug Release Profile

Release model: Weibull | Release order: Sustained first-order | Max EE: 75%

Metric	Blood (pH 7.4)	Endosomal (pH 5.5)	Significance
t50 (50% release)	21.0 h	42.2 h	Faster endo = better escape
t90 (90% release)	48.0 h	N/A	Complete release window
Rate constant k	0.0300 /h	0.0150 /h	Higher endo = pH-responsive
Max released	75%	75%	= encapsulation efficiency

5. Shelf-Life & Degradation Predictor

Grade: MARGINAL (3-6 months) | t90 = 105 days | Dominant pathway: Drug leakage | Storage: 4 degC

Pathway	Rate (k, /day)	Relative %
Hydrolysis	0.000202	20%
Oxidation	0.000172	17%
Aggregation	0.000265	26%
Drug leakage	0.000368	37%

6. Colloidal Stability — DLVO Theory + Protein Corona

DLVO potential energy $V_{\text{total}} = 8.9 \text{ kT}$ (UNSTABLE ~~x~~ — REFORMULATE) | Threshold: $>25 \text{ kT}$ prevents aggregation in blood. Reference: Verwey & Overbeek 1948; Derjaguin 1987.

Parameter	Value	Equation	Significance
DLVO V_{total} (kT)	8.9	$V_{\text{vdW}} + V_{\text{EDL}}$	$>25\text{kT} = \text{stable}$
V_{vdW} (attraction)	N/A	$-A \cdot R / (12h)$	Hamaker constant
V_{EDL} (repulsion)	N/A	$64\pi \epsilon R \gamma^2 e^{-\kappa h}$	Zeta potential
Debye length (nm)	N/A	$1/\kappa$	Ionic screening
Protein corona (nm)	N/A	Langmuir model	Stealth reduction
Opsonin index	N/A	IgG + C3b binding	$<0.3 = \text{good}$
Stealth index	N/A	PEG brush model	$0.6+ = \text{good}$
MPS clearance (h)	N/A	Michaelis-Menten	$>12\text{h}$ adequate

7. Nanotoxicity & Immunogenicity Screening

Immunogenicity grade: **RISK -- REFORMULATE** | Score: 52/100

Test	Score / Result	Risk	Threshold
CARPA (complement)	0.982	HIGH	$>0.6 = \text{HIGH}$
Anti-PEG antibody	0.250	LOW	$0.5+ = \text{caution}$
MPS uptake	0.473	—	$>0.6 = \text{rapid clearance}$
Cytokine storm	—	LOW	Charge-dependent
Platelet activation	—	LOW	Cationic $>200\text{nm}$

Mitigations:

→ Consider pre-medication with antihistamines (H1/H2 blockers)

8. Off-Target Toxicity — 50-Receptor QSAR Panel

Overall: HIGH RISK -- 8 off-target hits | Cardiac risk: NO ✓ | Hepatic risk: NO ✓ | CNS off-target: YES ■

Receptor	Free Drug Score	In-DDS Score	Risk
hERG_K+	0.000	0.000	LOW
Nav1.5	0.000	0.000	LOW
Cav1.2	0.000	0.000	LOW
CYP3A4_inhib	0.000	0.000	LOW
CYP2D6_inhib	0.955	0.597	HIGH
CYP2C9_inhib	0.173	0.108	LOW
CYP1A2_inhib	1.000	0.625	HIGH
CYP2C8_inhib	0.000	0.000	LOW
OATP1B1	0.079	0.049	LOW
MDR1_Pgp	0.000	0.000	LOW
BCRP	0.000	0.000	LOW
MRP2	0.000	0.000	LOW
DAT_dopamine	1.000	0.625	HIGH
SERT_serotonin	1.000	0.625	HIGH
5HT2A	1.000	0.625	HIGH
D2_dopamine	1.000	0.625	HIGH
GABA_A	0.328	0.205	LOW
NMDA	0.898	0.561	HIGH
sigma1	1.000	0.625	HIGH
ERalpha_estrogen	0.000	0.000	LOW

High-risk flags (8):

- CYP2D6_inhib: HIGH (score=0.60)
- CYP1A2_inhib: HIGH (score=0.62)
- DAT_dopamine: HIGH (score=0.62)
- SERT_serotonin: HIGH (score=0.62)
- 5HT2A: HIGH (score=0.62)
- D2_dopamine: HIGH (score=0.62)
- NMDA: HIGH (score=0.56)
- sigma1: HIGH (score=0.62)

9. Advanced Science Modules — Key Outputs

9a. Competitive DDS Landscape (ClinicalTrials.gov)

Our DDS: BBB Score = 100 | Position: SUPERIOR | Active trials found: 3

NCT ID	Title	Phase	N
NCT06846658	Exploring the Olfactory Mucosa, Blood and Uri	['?']	180
NCT06855368	Biomarkers for Monitoring Dementia Progressio	['?']	300
NCT04899908	Stereotactic Brain-directed Radiation With or	['PHASE2']	134

9b. Patient Subgroup Stratification

Overall response: 33.4% | Best: Very elderly (>80) (37%) | Worst: Young adults (18-40) (29%) | Recommendation:

Highest efficacy in Very elderly (>80) (predicted response 37%). Dose reduce by

Subgroup	Pop. Freq.	CNS Bioavail	Response%	Tox Risk	Dose Adj.
Young adults (18-40)	20% of patients	8.3%	29.3%	LOW	x1.29 standard dose
Middle-aged (40-65)	35% of patients	9.5%	33.5%	LOW	x1.14 standard dose
Elderly (65-80)	30% of patients	9.7%	34.3%	LOW	x0.72 standard dose
Very elderly (>80)	15% of patients	10.5%	36.9%	MODERATE	x0.46 standard dose

9d. Lysosomal Trafficking Predictor

Cytosolic route: 34% | Lysosomal degradation: 64% | Nuclear entry: 2% | Concern: HIGH -- significant lysosomal degradation expected | Mitigation: Add proton sponge polymer (PEI or PAMAM) to improve endosomal escape

9e. LNP Ionization State

Estimated pKa = 5.8 | Endosomal escape prediction = 51% | Optimal ionizable lipid pKa = 5.8 (suboptimal -- target pKa 6.0-6.8 for endosoma

10. Synthetic Clinical Trial — N=500 Virtual Patients

Monte Carlo simulation of Phase 1 trial. Patient covariates: age, weight, CYP3A4 genotype, renal function, BBB integrity by age. Pharmacokinetics computed per-patient from PBPK model. Reference: FDA PBPK guidance 2018; Lalonde 2007.

NO-GO / REFORMULATE Go/No-Go	26.0% Response Rate	0.0% Severe AE	1.07 mg/kg Optimal Dose	500 N Patients
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Endpoint	Result	Benchmark	Status
Overall response rate	26.0%	>60% = GO	✗
Severe AE rate	0.0%	<5% = GO	✓
Young adults (<65)	29.4%	—	—
Elderly (>65)	24.9%	—	—
Mild AE	0.4%	—	—
Renal AE	0.0%	<15%	✓

Recommendation: Predicted Phase 1 success rate: 26%. Dose 1.07 mg/kg recommended. Monitor renal function in 0% of elderly patients.

11. Manufacturing & Sterilization Modules

11a. Terminal Sterilization Survivability

Recommended method: Gamma 25kGy | Cost implication: Gamma or E-beam (~3x cost) | Feasible methods: gamma 25kGy, aseptic filtration 022um, VHP H2O2

Method	Survives?	Detail
autoclave 121C	NO ✗	Tm=42 degC < 121 degC -- LIPID MELTS
gamma 25kGy	YES ✓	Survives gamma sterilization
aseptic filtration 022um	YES ✓	Size 105 nm < 220 nm -- filterable
VHP H2O2	YES ✓	PEG provides surface protection

11b. Lyophilization Cycle Optimizer

Parameter	Value	Standard
Tg' (cryo.	-30°C	Formulation-specific
Primary drying T	-32.0°C	Must be < Tg'
Primary P	0.1478 mbar	0.5×P_ice
Primary time	25.6 h	Pikal 2002 model
Secondary T	25.0°C	+25°C
Cycle total	41.2 h	Include ramp time
Cake collapse	OK: Tg within safe margin	Must say OK

11c. Cryo-Chain Excursion Predictor

Excursion: -20.0°C for 4.0h | EE: 75.0% → 75.0% (loss: 0.0%) | Decision: RELEASE BATCH | Confidence: 98%

11d. Adversarial Stress-Testing

Grade: 2/5 scenarios passed -- CONDITIONAL | 2/5 scenarios passed.

Scenario	Pass/Fail	Detail
Fever 40C	FAIL ✗	Tm=42.0degC -- RISK: may partially melt at fever temperature
Shear Stress	PASS ✓	Carrier withstands capillary shear
Oxidative Stress	FAIL ✗	Lipid oxidation t1/2 = 35h (RISK: add antioxidant like alpha-tocophero
High Protein Plasma	FAIL ✗	Stealth index 0.50: RISK: rapid clearance in protein-rich plasma
Repeated Dosing	PASS ✓	Anti-PEG IgM may reduce t1/2 to 70% of first dose after re-dosing. Mon

12. Regulatory Compliance & Supply Chain

12a. FDA 21 CFR Part 11 Compliance

Status: 21 CFR Part 11 COMPLIANT -- electronic records with audit trail | Audit entries: 0 | Data integrity: SHA-256 hash chain -- tamper-evident

12b. Geopolitical Supply-Chain Risk Assessment

Overall supply risk: HIGH | Supply chain score: 67/100 | Recommendation: Establish 6-month buffer stock for high-risk materials

Material	Risk	Source	HHI Index	Mitigation
DPPC	MODERATE	Multiple (Lipoid, NOF)	0.3	Maintain 6-month safety stock
Cholesterol	LOW	Multiple commodity	0.1	Maintain 6-month safety stock
DSPE-PEG2000	HIGH	Single (NOF Corp, Japan)	0.7	Qualify alternative supplier

12c. Digital Pharmacovigilance — Post-Elimination Fate

Unchanged excretion: 20.7% | Eco-persistence: 0.5 days | Environmental risk: LOW | Renal dose adjustment: Reduce by 50%

13. Literature Citations — PubMed Auto-Mined

Top relevant publications retrieved live from PubMed E-utilities API based on drug name + carrier type. These papers confirm or contextualize the computational predictions above.

[1] Shi J et al. (2017). Cancer nanomedicine... Nat Rev Cancer. PMID:30291251

14. Biophysical Binding & CNS-Specific Modules

14a. FEP+ Binding Affinity (LIE approximation)

Ligand: transferrin | $\Delta G_{\text{single}} = -11.00$ kcal/mol | $\Delta G_{\text{avidity}} = -8.59$ kcal/mol | $K_d = 885$ nM (Weak (>100 nM)) | $n_{\text{ligands}} = 27709$ | Residence time = 1129575.8 s

14b. Glymphatic Clearance Simulation

ECM binding index = 0.740 | $t_{1/2}$ waking = 12.4 h | t_{90} clearance = 41.4 h | GOOD: particle size > ECM pore -- extended CNS retention

14c. Microglial Activation & Neuroinflammation

Neuroinflammation score = 0.104 | Risk: LOW -- safe for CNS application | IL-6 fold-change = 2.0 $\times$ | TNF- α = 1.8 $\times$
→ Increase PEGylation to 5-10 mol% for complement evasion

14d. FUS-Responsive Nanocarrier Design

Frequency = 0.5 MHz | MI target = 0.40 | BBB open window = 42 min | FUS uptake = 58.0% | Inject 105 nm DDS 5 min before FUS. FUS opens BBB for 42 min. Predicted 58% enha

14e. 62-Principle C+ Flow — Class A Surrogate (Top-1 DDS)

Top-1 DDS: Tf-PEG-Liposome **Composite:** 63.7/100 **Verdict:** ACCEPTABLE

Tf-PEG-Liposome: composite CNS-principle score = 63.7/100 (ACCEPTABLE). Carrier: liposome, size 0.0 nm, ζ +0.0 mV, ligand: Transferrin. Strongest group: G4 Safety (72.5/100); weakest: G8 Translational (0.0/100). Evaluated against all 57 Class A surrogate principles.

CNS Principle Group Rollups

CNS Group	Score (0-100)
G1 CNS Delivery	50.0
G2 Release Kinetics	45.5
G3 Stability	61.0
G4 Safety	72.5
G5 Glymphatic BBB	65.0
G6 Manufacturability	56.0
G7 DrugDDS Fit	68.7
G8 Translational	0.0

Top Strengths (Class A Surrogate)

Principle	Score	Method	Reference
P03	100.0	Novelty = 100 - frequency_of_combo_in_CNS_trials,	ClinicalTrials.gov landscape
P05	100.0	% subgroups responding (lower CYP risk = more univ	Whirl-Carrillo M et al (2012) Clin
P07	100.0	log-count proxy of literature support	NCBI E-utilities — heuristic (no li

Weak Spots (Below 60)

Principle	Score	Method	Reference
P44	4.8	AUC over 24h therapeutic window proxy	Bies RR et al (2019) Annu Rev Pharm
P13	3.4	AUC_brain/AUC_plasma proxy; 3-compartment surrogate	Hammarlund-Udenaes M et al (2008) P
P31	2.0	brain% = BBB · ligand · size_match · 100; 100 if ≥	Wilhelm S et al (2016) Nat Rev Mate

14e-ii. Complete 62-Principle Scoreboard (P01 → P62)

All 62 principles in canonical order, covering Class A (Surrogate), Class B (Deep Physics), and Class C (Translational).

P#	Cls	Title	Score	Conf.	Method
P01	A	Adversarial Stress-Testing Engine	68.3	MODERATE	Mean of 6 stress scenarios (pH4.5, pH2, pH8.5, 42°
P02	A	Cross-Species PK Scaling (Allometric)	60.0	MODERATE	EBW^0.75 scaling + class-specific adjustment
P03	A	Competitive DDS Landscape Radar	100.0	LOW	Novelty = 100 - frequency_of_combo_in_CNS_trials,
P04	A	Quantum Coherence Transport Model	50.0	LOW	MW ≥ 500 Da → tunneling negligible
P05	A	In-Silico Patient Subgroup Stratifier	100.0	MODERATE	% subgroups responding (lower CYP risk = more univ
P06	A	Lysosomal Trafficking Predictor	50.0	MODERATE	ECarrier endosomal_escape ⊕ drug-passive permeation
P07	A	Real-Time Literature Mining (PubMed)	100.0	LOW	log-count proxy of literature support
P08	A	Degradation Kinetics Under Oxidative S	30.0	MODERATE	EArrhenius k=A.exp(-Ea/RT) for carrier; minus penal
P09	A	Digital Pharmacovigilance Engine	85.0	MODERATE	EL00 - sum(metabolite organ accumulation risks)
P10	A	LNP Ionization State Predictor	63.7	MODERATE	ELipid Henderson-Hasselbalch ionization at endosoma
P11	A	Formulation Instability Fingerprint	100.0	MODERATE	EL00 - sum(weak-bond penalties from SMILES)
P12	A	CNS Disease-Stage-Aware Dosing	87.2	MODERATE	EScore = 100·BBB_integrity_factor × (0.4 + 0.6·CNS-
P13	A	PBPK Digital Twin	3.4	MODERATE	Eauc_brain/auc_plasma proxy; 3-compartment surrogat
P14	A	In-silico Dissolution & Release Profil	43.1	MODERATE	ES50 from carrier kinetics × drug-membrane partitio
P15	A	Shelf-life & Degradation Predictor	46.2	MODERATE	EArrhenius baseline × (0.7 + 0.3·EE) × drug_hydroly
P16	A	Scale-up & Manufacturability	20.0	MODERATE	EScale-readiness × shear-robustness, minus drug-pro
P17	A	Nanotoxicity & Immunogenicity Screenin	20.5	MODERATE	EMean of (hemolysis penalized by drug net+ charge,
P18	A	Active Targeting & Receptor Binding	74.2	HIGH	ELigand-affinity table × density × drug-compatibili
P19	A	QbD - Quality by Design Engine	100.0	MODERATE	EFraction of CQAs (size, zeta, EE, PDI, drug-loadin
P20	A	Cost-Efficiency Engine	0.0	LOW	Score = 100 - 3·(\$/mg estimate, drug+carrier+ligan
P21	C	Pre-IND Regulatory Reports	0.0	—	skipped_deep_validation_insufficient
P22	A	Protein Corona Predictor	2.0	MODERATE	EL00 × exp(-corona_thickness/10); thickness from ca
P23	A	Dynamic Crystal Polymorphism	98.0	MODERATE	EL00 - polymorph risk (rot_bonds + H-bond pattern)
P24	A	Shear-Stress & Scale-Up Collapse	14.3	MODERATE	Elog10(crit_shear/operating_shear) × 30; crit_shear
P25	A	Extractables & Leachables	100.0	MODERATE	EL00 - leachables risk by drug LogP × carrier
P26	A	Microbiome-Excipient Interactions	85.0	MODERATE	EL00 - mean_microbiome_degradability(carrier) - dru
P27	A	Lyophilization Cycle Optimizer	55.0	MODERATE	EL00 - (Tg' - 5) × 1.5
P28	A	3D-Printed Polypill Rheology	40.0	MODERATE	ECarrier-specific printability index
P29	A	Biomimetic & Exosome Engineering	65.0	MODERATE	EStealth-from-macrophage score; PEG ≥5% boost
P30	A	QM/MM Stimuli-Responsive Cleavage	45.0	MODERATE	EScore = 50 × 7.4 - pH_trigger
P31	A	In-Silico Biodistribution	2.0	MODERATE	Ebrain% = BBB · ligand · size_match · 100; 100 if ≥
P32	C	Automated FTO & IP Evader	0.0	—	skipped_deep_validation_insufficient
P33	A	BBB Quantum Breaker (Trojan-Horse Desi	58.2	MODERATE	E(0.6·ligand + 0.4·size) × density_factor

P#	Cls	Title	Score	Conf.	Method
P34	A	DNA Logic Gates & Bio-computing	15.0	MODERATE	DNA-based carriers score high
P35	A	Microgravity Formulation Engine	100.0	LOW	Sedimentation-Péclet proxy; smaller = better
P36	A	Geopolitical Supply-Chain Resilience	85.0	MODERATE	Supplier-diversity index by carrier + ligand
P37	A	Eco-Destructible Pharma	90.0	MODERATE	Biodegradability index by carrier class
P38	A	Glymphatic Clearance Trap	100.0	MODERATE	Stokes-Einstein: optimum 80-150 nm
P39	A	Microglial Activation & Neuroinflammat	100.0	MODERATE	EL00 - (cationic_charge + carrier_risk - PEG_protec
P40	A	Intranasal-to-Brain Delivery	55.0	MODERATE	Mucoadhesion + thermo-responsive boost
P41	A	Exosome Cargo Loading Thermodynamics	40.0	LOW	Not an exosome carrier — N/A
P42	A	Region-Specific Spatiotemporal Navigat	70.0	MODERATE	Ligand-region specificity table
P43	A	FUS-Responsive Nanocarriers	40.0	MODERATE	FUS-response by carrier acoustic properties
P44	A	CNS-Specific PBPK Time-Machine	4.8	MODERATE	AUC over 24h therapeutic window proxy
P45	C	FDA 21 CFR Part 11 Compliance	0.0	—	skipped_deep_validation_insufficient
P46	A	Polypharmacy & DDI Simulator	100.0	MODERATE	CYP-inhibition heuristic from SMILES
P47	B	Free Energy Perturbation (FEP+)	72.0	LOW	Vina-like ΔG proxy from MW + LogP (deep mode runs
P48	A	Off-Target Toxicity & QSAR (50-recepto	100.0	MODERATE	50-receptor QSAR surrogate (hERG, 5HT2B, AhR risks
P49	A	Organ-on-a-Chip Simulator	80.0	MODERATE	Microfluidic compatibility (size + PDI)
P50	A	Cryo-Chain Thermal Excursion Predictor	71.5	MODERATE	Distance from -20°C phase transition × 1.3
P51	A	Terminal Sterilization Survivability	50.0	MODERATE	Carrier gamma-radiation survival (25 kGy)
P52	A	Continuous Manufacturing Digital Twin	85.0	MODERATE	Continuous-process readiness by carrier
P53	A	Dark Data & Negative Results Vault	100.0	MODERATE	Similarity-to-documented-failure
P54	A	Pharmacogenomic-Guided Targeting	90.0	MODERATE	Class-based pharmacogenomic applicability
P55	C	Automated Grant & NIH Proposal Generat	0.0	—	skipped_deep_validation_insufficient
P56	C	Patentability Score Engine	0.0	—	skipped_deep_validation_insufficient
P57	A	Microfluidics & LNP Synthesis Digital	90.0	MODERATE	Microfluidic readiness for size 50-150 nm
P58	A	Impurity Cascade Predictor	90.0	MODERATE	Carrier residual-metals impurity profile
P59	A	4D Shape-Shifting Carriers	30.0	MODERATE	Stimuli-responsive bonus by release kinetics
P60	A	Swarm Nanorobotics Intelligence	15.0	LOW	Swarm capability (most carriers: no)
P61	A	Synthetic Clinical Trials & Virtual Hu	30.6	MODERATE	Virtual-cohort responder fraction
P62	A	Biobetter / Supergeneric Generator	40.0	LOW	Novelty distance from on-market CNS DDS

62 principles evaluated. Composite: 63.7/100 (ACCEPTABLE)

14f. Class B Deep Physics Validation (Top-1 DDS)

Verdict: **FAILED** **Pass rate:** 42.9% (12/28 principles validated, threshold 70%)

Combined score (surrogate pass-through + real physics): 12/28 principles scored PASSED (42.9%). Of these, only 7/28 principles ran independent deep computation (passed 3/7 = 42.9%); the rest re-used their Class-A surrogate score pending a future full-physics HPC run. Cite independent_pct, not pct, as evidence of physics-based validation.

Per-principle deep validation

P#	Validated	Score	Value	Conf.	Method
P01	✓	68.3	68.33	MODERATE	Full MD stress simulation — surrogate value confirmed (
P02	✓	65.0	2444.16	LOW	Multi-parameter allometric scaling (Mahmood 2007): half
P04	✗	50.0	50	LOW	Full QM tunneling (QCElemental + ASE) — surrogate value
P08	✗	30.0	30.0	LOW	Full radical-chain reaction MD — surrogate value confir
P10	✓	63.7	63.68	MODERATE	Constant-pH MD simulation — surrogate value confirmed (
P11	✓	100.0	100	MODERATE	DFT bond dissociation energies — surrogate value confir
P12	✓	87.2	87.25	MODERATE	Stage-specific PBPK with full BBB physiology — surrogat
P13	✗	23.0	0.0115	MODERATE	3-compartment PBPK ODE (scipy.integrate.odeint): blood
P16	✗	20.0	20	LOW	CFD of 1000L bioreactor — surrogate value confirmed (fu
P18	✓	100.0	-10.5	HIGH	MM/GBSA-style ΔG estimate from validated ligand-recepto
P23	✓	98.0	98.0	MODERATE	CSP via DFT — surrogate value confirmed (full-physics H
P24	✗	14.3	14.31	LOW	Full CFD + MD coupling — surrogate value confirmed (ful
P29	✓	65.0	65	MODERATE	Full membrane-fusion MD — surrogate value confirmed (fu
P30	✗	45.0	45.0	LOW	Full QM/MM (PySCF or ORCA) — surrogate value confirmed
P31	✗	100.0	0.1	MODERATE	7-organ whole-body distribution with size + charge + li
P33	✗	58.2	58.2	LOW	Atomistic docking + SMD pulling — surrogate value confi
P38	✓	99.2	12.6	HIGH	Stokes-Einstein $D = k_B \cdot T / (6\pi\eta r)$ at 37°C, $\eta_{\text{CSF}} = 7 \times 10^{-4}$ Pa·s
P40	✗	55.0	55	LOW	Full nasal cavity CFD — surrogate value confirmed (full
P41	✗	40.0	40	LOW	Full membrane mechanics MD — surrogate value confirmed
P43	✗	40.0	40	LOW	Full acoustic radiation force MD — surrogate value conf
P44	✗	0.0	0.0	MODERATE	4-compartment CNS-PBPK ODE with glymphatic clearance: B
P47	✗	66.5	-6.05	MODERATE	Enhanced FEP+: Vina baseline + hydration correction (Lo
P50	✓	71.5	71.5	MODERATE	Full coarse-grained MD lipid phase transition — surroga
P51	✗	50.0	50	LOW	Full radical-chain damage MD — surrogate value confirme
P57	✓	90.0	90	MODERATE	Full CFD of mixer geometry — surrogate value confirmed
P58	✓	90.0	90	MODERATE	Full QM/MM cascade simulation — surrogate value confirm
P59	✗	30.0	30	LOW	Full coarse-grained MD morphological transition — surro
P61	✗	30.6	31.0	LOW	Full Monte-Carlo population PBPK — surrogate value conf

14g. Class C Translational Deliverables

Principle	Title	Status	Score / Outcome
P21		skipped_deep_validation_	—
P32		skipped_deep_validation_	—
P45		skipped_deep_validation_	—
P55		skipped_deep_validation_	—
P56		skipped_deep_validation_	—

Narratives

P21: Deep validation passed only 12/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P32: Deep validation passed only 12/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P45: Deep validation passed only 12/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P55: Deep validation passed only 12/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P56: Deep validation passed only 12/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

v23: structured outlines for Pre-IND, Grant, Patent will be rendered as fully formatted Word/PDF deliverables.

14h. Top-N Fallback Audit Trail

If the Top-1 DDS fails deep validation ($\geq 70\%$ threshold), the orchestrator falls back to Top-2, then Top-3. Each candidate's outcome and the explicit transition reasoning are recorded below.

Rank	DDS Name	Verdict	Pass %	Promoted?
#1	Tf-PEG-Liposome	FAILED	42.9%	—
#2	RVG29-Liposome-pH	FAILED	42.9%	—
#3	AAV9-vector	FAILED	35.7%	—

#1 Tf-PEG-Liposome — FAILED

Failure reason: Deep validation FAILED: only 12/28 principles validated (42.9%, threshold 70%). Critical failures: P04, P08, P13, P16, P24.

Transition reason: Falling back to rank #2 because deep physics showed insufficient evidence for this DDS.

Failed principles: P04, P08, P13, P16, P24, P30, P31, P33

#2 RVG29-Liposome-pH — FAILED

Failure reason: Deep validation FAILED: only 12/28 principles validated (42.9%, threshold 70%). Critical failures: P04, P08, P13, P16, P24.

Transition reason: Falling back to rank #3 because deep physics showed insufficient evidence for this DDS.

Failed principles: P04, P08, P13, P16, P24, P30, P31, P33

#3 AAV9-vector — FAILED

Failure reason: Deep validation FAILED: only 10/28 principles validated (35.7%, threshold 70%). Critical failures: P04, P08, P13, P16, P18.

Transition reason: No more candidates in Top-3 — reverting to rank #1 and reporting with FAILED verdict. Researcher should reformulate.

Failed principles: P04, P08, P13, P16, P18, P24, P30, P31

15. Executive Decision Framework

Criterion	Status	Evidence
DLVO colloidal stability	FAIL ✗	V_total = 9 kT
BBB penetration >10%	FAIL ✗	DDS BBB = N/A%
Cardiac off-target	PASS ✓	hERG/Nav/Cav QSAR
Clinical trial GO	NO-GO / REFORMULATE	Response 26% AE 0.0%
Sterilization feasible	3 method(s)	Gamma 25Kgy
Supply chain	HIGH	Score 67/100
OVERALL DECISION	CONDITIONAL GO	Proceed to IND-enabling studies

NOTE: All results are computational predictions. Required wet-lab validation: DLS/Zeta (physical stability), HPLC/NMR (drug content), BBB TEER assay, in vivo PK in rodent model.